CSA 0712 - Computer Networks

CO4AT3 - Comparative Analysis

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Date : 23/08/26

1.

No	Multiplexing	Demultiplexing
1.	Happens at the sender side	Happens at the receiver side
2.	Collects data from multiple Application	Receives segments from the network
3.	Combines data for transmission	Separates received data
4.	uses application sockets	identifies the Correct destination socket
5.	Adds source port numbers	checks destination port numbers.
6.	Adds destination port numbers.	uses port information for delivery
7.	Encapsulates data into segments	Extracts data from segments
8.	Allows many applications to share transport services	Sends data to the Correct application
9.	Mainly prepares data for sending	Mainly delivers data after receiving
10.	Eg: Multiple apps sending data simultaneously	Eg: Browser and email receiving their own data

2.

No	TCP sockets	UDP Datagrams
1.	Connection - oriented	Connectionless
2.	Provides reliable communication	Does not guarantee reliability
3.	Stream - based Communication	Message - based Communication
4.	Maintains Connection state	Stateless Communication
5.	uses acknowledgments	No acknowledgments by default.
6.	Ensures ordered delivery	Packets may arrive out of order
7.	Performs error recovery	No automatic recovery
8.	Slower due to reliability features	faster with less overhead
9.	uses flow and Congestion Control	Does not provide built in control
10.	used for web pages and file transfer	used for streaming, gaming and DNS.

3.

No	Multicast	Unicast / Broadcast
1.	Sends data to a selected group	unicast sends to one; broadcast sends to all.
2.	uses one stream for multiple receivers	May require multiple Copies or reach everyone
3.	usually uses UDP	Can use different Communication methods
4.	only subscribed hosts receive multicast data	unicast targets one host; broadcast targets all hosts
5.	Efficient for group Communication	Can Create more network overhead.
6.	Saves bandwidth	May Consume more bandwidth
7.	suitable for live streaming	unicast is suitable for one-to-one Communication.
8.	uses multicast group addresses	uses individual or broadcast addresses
9.	Reduces duplicate transmissions	unicast may duplicate the same data many times
10.	Eg: online live video	Eg: one to one message

No	TCP Congestion Control	Traffic Management.
1.	Managed by TCP endpoints	often managed by network devices
2.	Controls sending rate	Controls network traffic flow.
3.	Detects Congestion using packet loss	uses router and network policies
4.	uses algorithms like AIMD	uses techniques like QOS and RED
5.	uses slow start	uses queue management
6.	Reduces speed during Congestion	Prioritizes or limits traffic
7.	works end to end	works inside the network
8.	Reacts after detecting Congestion	Can proactively manage traffic
9.	focuses on TCP connections	Can manage different types of traffic
10.	Eg: TCP reduces congestion window	eg: Router drops.

5.

No	M/M/1 Single - Server	M/M/c Multi - Server
1.	Has one server	Has multiple servers
2.	One Customer is served at a time	Multiple Customers can be served.
3.	Arrival follows Poisson Process	Arrival also Commonly follows a Poisson
4.	Service time is exponentially distributed	Service time is exponentially distribute
5.	Traffic intensity is $\rho = \lambda/\mu$	Traffic intensity is $\rho = \lambda/(c\mu)$
6.	Requires $\rho < 1$ for stability	Requires $\rho < 1$ for stability
7.	Average waiting can become high during heavy traffic	Multiple server reduce waiting time
8.	Simple and low Cost model	More Complex and Costly model
9.	Suitable for a single service Counter	suitable for multiple service Counters
10.	Eg: one ATM machine	Eg: Multiple bank Counters

6.

No	Priority Queues	Network of Queues
1.	Packets are assigned priorities	Multiple queues are Connected together.
2.	high - priority packets are served first	Packets move between different queue nodes
3.	Service order depends on priority	service order depends on network flow
4.	important traffic gets faster service	traffic may pass through several queues
5.	Low - priority traffic may wait longer	waiting can occur at multiple stages
6.	used for QoS	used to model Complex systems.
7.	focuses on packet importance.	focuses on packet movement between nodes
8.	Eg: voice traffic	Eg. Packets moving through multiple routers.